

Week 9 Tutorial Questions

PHYS3114

Problem 9.1: [Griffiths 4th edition problem 12.26] Find the invariant product of the 4-velocity with itself $\eta^\mu \eta_\mu$.

Problem 9.2: [Griffiths 4th edition problem 12.2/12.28]
Consider a particle in hyperbolic motion,

$$x(t) = \sqrt{b^2 + (ct)^2}, \quad y = z = 0$$

- (a) Find the proper time τ as a function of t , assuming the clocks are set so that $\tau = 0$ when $t = 0$.
- (b) Find x and v (ordinary velocity) as functions of τ .
- (c) Find η^μ (proper velocity) as a function of t .

Problem 9.3: [Griffiths 4th edition problem 12.33]

A particle of mass m whose total energy is twice its rest energy collides with an identical particle at rest. If they stick together, what is the mass of the resulting composite particle? What is its velocity?

Problem 9.4: [Griffiths 4th edition problem 12.34]

A neutral pion of (rest) mass m and (relativistic) momentum $p = \frac{3}{4}mc$ decays into two photons. One of the photons is emitted in the same direction as the original pion, and the other in the opposite direction. Find the (relativistic) energy of each photon.

Problem 9.5: [Griffiths 4th edition problem 12.36]

In a pair annihilation experiment, an electron (mass m) with momentum p_e hits a positron (same mass, but opposite charge) at rest. They annihilate, producing two photons. (Why couldn't they produce just one photon?) If one of the photons emerges at 60° to the incident electron direction, what is its energy?